

```
In [1]: name = "John"      # String
        age = 25          # Integer
        height = 5.9      # Float
        is_student = True # Boolean

        print(name, age, height, is_student)
```

John 25 5.9 True

```
In [2]: name = input("Enter your name: ")
        print("Hello", name)
```

Enter your name: John
Hello John

```
In [3]: age = int(input("Enter your age: "))
        if age >= 18:
            print("You are an adult.")
        else:
            print("You are a minor.")
```

Enter your age: 17
You are a minor.

```
In [4]: for i in range(5):
        print("Loop number:", i)
```

Loop number: 0
Loop number: 1
Loop number: 2
Loop number: 3
Loop number: 4

```
In [5]: fruits = ["apple", "banana", "cherry"]
        for fruit in fruits:
            print(fruit)
```

apple
banana
cherry

```
In [6]: a = 10
        b = 5
        sum = a + b
        print("Sum is:", sum)
```

Sum is: 15

```
In [7]: num = int(input("Enter a number: "))
        if num % 2 == 0:
            print("Even number")
        else:
            print("Odd number")
```

Enter a number: 6
Even number

```
In [8]: colors = ["red", "green", "blue"]
        for color in colors:
            print("Color:", color)
```

Color: red
Color: green
Color: blue

```
In [9]: for i in range(1, 11):
        print("3 x", i, "=", 3 * i)
```

3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30

```
In [10]: a = 45
         b = 78
         c = 32

         if a >= b and a >= c:
             print("Largest:", a)
         elif b >= a and b >= c:
             print("Largest:", b)
         else:
             print("Largest:", c)
```

Largest: 78

```
In [11]: text = "education"
         vowels = "aeiou"
         count = 0

         for char in text:
             if char in vowels:
                 count += 1

         print("Number of vowels in", text, "is", count)
```

Number of vowels in education is 5

```
In [12]: x = 10
         print("Initial x:", x)

         x += 5      # x = 10 + 5
         print("After x += 5:", x)

         x -= 3      # x = 15 - 3
         print("After x -= 3:", x)

         x *= 2      # x = 12 * 2
         print("After x *= 2:", x)

         x /= 4      # x = 24 / 4
```

```
print("After x /= 4:", x)

x %= 5      # x = 6 % 5
print("After x %= 5:", x)

x **= 3     # x = 1 ** 3
print("After x **= 3:", x)

x //= 2     # x = 1 // 2
print("After x //= 2:", x)
```

Initial x: 10
After x += 5: 15
After x -= 3: 12
After x *= 2: 24
After x /= 4: 6.0
After x %= 5: 1.0
After x **= 3: 1.0
After x //= 2: 0.0

In [13]:

```
for i in range(5): # 0 to 4
    print(i)
```

0
1
2
3
4

In [14]:

```
for i in range(1, 6): # Start at 1, end before 6
    print(i)
```

1
2
3
4
5

In [15]:

```
for i in range(2, 11, 2): # Step of 2
    print(i)
```

2
4
6
8
10

In [16]:

```
for i in range(5, 0, -1): # Start at 5, end before 0, step -1
    print(i)
```

5
4
3
2
1

In [17]:

```
for i in range(3, 31, 3):
    print(i)
```

3
6

9
12
15
18
21
24
27
30

```
In [18]: for i in range(1, 11):  
         print("Natural number:", i)
```

Natural number: 1
Natural number: 2
Natural number: 3
Natural number: 4
Natural number: 5
Natural number: 6
Natural number: 7
Natural number: 8
Natural number: 9
Natural number: 10

```
In [19]: for i in range(1, 6):  
         print("Square of", i, "is", i * i)
```

Square of 1 is 1
Square of 2 is 4
Square of 3 is 9
Square of 4 is 16
Square of 5 is 25

```
In [20]: for i in range(1, 6):  
         print("*" * i)
```

*
**


```
In [21]: numbers = [1, 2, 3, 4, 5]  
  
         for num in numbers:  
             print("Square of", num, "is", num ** 2)
```

Square of 1 is 1
Square of 2 is 4
Square of 3 is 9
Square of 4 is 16
Square of 5 is 25

```
In [22]: nums = [10, 20, 30, 40]  
         total = 0  
  
         for num in nums:  
             total += num  
  
         print("Total sum:", total)
```

Total sum: 100

```
In [23]: even_numbers = list(range(2, 21, 2))
print("Even numbers:", even_numbers)
```

Even numbers: [2, 4, 6, 8, 10, 12, 14, 16, 18, 20]

```
In [24]: squares = []

for i in range(1, 6):
    squares.append(i ** 2)

print("Squares:", squares)
```

Squares: [1, 4, 9, 16, 25]

```
In [25]: marks = [45, 67, 89, 34, 76]

for mark in marks:
    if mark > 50:
        print("Pass:", mark)
```

Pass: 67
Pass: 89
Pass: 76

```
In [26]: names = ["Tom", "Jerry", "Spike"]

for i in range(len(names)):
    print("Index:", i, "Name:", names[i])
```

Index: 0 Name: Tom
Index: 1 Name: Jerry
Index: 2 Name: Spike

```
In [27]: multiples = []

for i in range(1, 11):
    multiples.append(i * 3)

print("Multiples of 3:", multiples)
```

Multiples of 3: [3, 6, 9, 12, 15, 18, 21, 24, 27, 30]

```
In [28]: fruits = ["apple", "banana", "cherry", "mango"]
fruits.reverse()
print("Reversed list:", fruits)
```

Reversed list: ['mango', 'cherry', 'banana', 'apple']

```
In [29]: numbers = [10, 20, 30, 40, 50, 60]

for i in range(0, len(numbers), 2): # Step = 2
    print("Jumped item:", numbers[i])
```

Jumped item: 10
Jumped item: 30
Jumped item: 50

```
In [30]: fruits = ["apple", "banana"]
         fruits.append("orange")
         print(fruits)
```

```
['apple', 'banana', 'orange']
```

```
In [31]: fruits.insert(1, "grapes") # Insert at index 1
         print(fruits)
```

```
['apple', 'grapes', 'banana', 'orange']
```

```
In [32]: fruits = ["apple", "banana", "orange"]
         fruits.remove("banana")
         print(fruits)
```

```
['apple', 'orange']
```

```
In [33]: del fruits[1]
         print(fruits)
```

```
['apple']
```

```
In [34]: last_fruit = fruits.pop() # Removes last item
         print("Removed:", last_fruit)
         print(fruits)
```

```
Removed: apple
[]
```

```
In [35]: fruits = ["apple", "banana", "cherry"]

         if "banana" in fruits:
             print("Yes, banana is in the list")
         else:
             print("No, banana not found")
```

```
Yes, banana is in the list
```

```
In [36]: marks = [10, 20, 10, 30, 10]
         print("Count of 10:", marks.count(10))
```

```
Count of 10: 3
```

```
In [37]: names = ["Alice", "Bob", "Charlie"]

         for i in range(len(names)):
             print(f"Index {i} = {names[i]}")
```

```
Index 0 = Alice
Index 1 = Bob
Index 2 = Charlie
```

```
In [38]: fruits = ["apple", "banana", "cherry"]
         fruits.clear()
         print("List after clearing:", fruits)
```

```
List after clearing: []
```

```
In [39]: list1 = [1, 2, 3]
list2 = [4, 5, 6]
combined = list1 + list2
print("Combined list:", combined)
```

Combined list: [1, 2, 3, 4, 5, 6]

```
In [40]: text = "Hello, World"
print(text[0:5])    # Hello
print(text[-5:])    # World
print(text[:2])     # Hlo ol
```

Hello
World
Hlo ol

```
In [41]: word = "Python"
for char in word:
    print(char)
```

P
y
t
h
o
n

```
In [42]: text = "Hello Python"
print(text.upper())    # HELLO PYTHON
print(text.lower())    # hello python
```

HELLO PYTHON
hello python

```
In [43]: text = "Techstack"
print("Length:", len(text)) # 9
```

Length: 9

```
In [44]: text = "Python is fun"
print("Python" in text)    # True
print("Java" not in text)  # True
```

True
True

```
In [45]: text = "I love Java"
new_text = text.replace("Java", "Python")
print(new_text)
```

I love Python

```
In [46]: first = "Hello"
second = "World"
combined = first + " " + second
print(combined)
```

Hello World

```
In [47]: sentence = "banana"
print("Count of a:", sentence.count('a')) # 3
```

Count of a: 3

```
In [48]: text = "Python"
print("Reversed:", text[::-1]) # nohtyP
```

Reversed: nohtyP

```
In [49]: text = "  Hello World  "
print("Trimmed:", text.strip()) # Removes spaces from both ends
```

Trimmed: Hello World

```
In [50]: text = "Python is amazing"
words = text.split() # By space by default
print(words) # ['Python', 'is', 'amazing']
```

['Python', 'is', 'amazing']

```
In [51]: words = ['Python', 'is', 'awesome']
sentence = " ".join(words)
print(sentence)
```

Python is awesome

```
In [53]: x = "1234"
y = "abc"

print(x.isdigit()) # True
print(y.isalpha()) # True
```

True

True

In []: